VALIDATION REPORT
OPERATIONAL QUALIFICATION
SHELF LIFE & DEGRADATION ANALYSIS MODULE

JR Validated Environment — Shelf Life & Degradation Analysis Module v1.0

Field	Value
Document Number	JR-VR-SHELF-001
Title	Validation Report — Operational Qualification, Shelf Life & Degradation Analysis Module
Validation Plan	JR-VP-SHELF-001 v1.0
System	JR Validated Environment — Shelf Life Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-04-19
OQ Result	PASS — 57 / 57 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-04-19	Joep Rous	Initial OQ execution report. All 57 test cases. 57/57 PASS.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-04-19
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Shelf Life & Degradation Analysis Module, version 1.0. It provides objective evidence that all Shelf Life scripts perform as specified in OQ Validation Plan JR-VP-SHELF-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 57 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_shelf_life_oq.

2. Scope

2.1 In Scope

- All 5 R scripts in repos/shelf_life/R/ (jrc_shelf_life_q10, jrc_shelf_life_arrhenius, jrc_shelf_life_linear, jrc_shelf_life_poolability, jrc_shelf_life_extrapolate)
- Correct computation of AF and real-time equivalent via Q10 (ASTM F1980) and Arrhenius (ISO 11607 / ICH Q1E) methods; numerical correctness verified by independent Python arithmetic
- Correct linear degradation modelling: Brown-Forsythe test, OLS regression, shelf life estimation as CI of predicted mean crossing the specification limit (ICH Q1E)
- Correct --transform log behaviour: log-scale OLS fit with CI back-transformed via exp()
- Correct ICH Q1E batch poolability: two-step ANCOVA at α=0.25; FULL/PARTIAL/DO NOT POOL decision
- Correct extrapolation with transform-aware CI back-transformation; ICH Q1E 50% warning and 100% hard stop
- Correct rejection of invalid inputs; bypass-protection verification (RENV_PATHS_ROOT check)
- PNG and model CSV output to ~/Downloads/

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real product stability data
- Validation of R interpreter and third-party packages (ggplot2, nlme)
- Statistical validation of the underlying methods (Arrhenius kinetics, ICH Q1E poolability rules, OLS inference) — see JR-VP-SHELF-001 Section 3 for references

3. Reference Documents

Document ID	Title	Location
JR-VP-SHELF-001 v1.0	Validation Plan — OQ, Shelf Life & Degradation Analysis Module	repos/shelf_life/docs/shelf_life_validation_plan.pdf
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	~/.jrscript/<PROJECT_ID>/validation/IQ_validation_*.txt
OQ Evidence	Shelf Life OQ Execution Evidence (pytest output + header)	shelf_life_oq_execution_20260419T080311.txt

4. Test Environment

Component	Details
Execution date / time	2026-04-19T08:03:11
Host	Joeps-MacBook-Air.local
Operating System	15.7.3
R Version	4.5.3
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/shelf_life/admin_shelf_life_oq
Test file	repos/shelf_life/oq/test_shelf_life_suite.py
Test data	repos/shelf_life/oq/data/ (13 CSV fixtures)
Execution command	bash repos/shelf_life/admin_shelf_life_oq
OQ evidence file	shelf_life_oq_execution_20260419T080311.txt

5. Test Results

All 57 test cases were executed on 2026-04-19. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

Script	Tests Executed	Result
jrc_shelf_life_q10	10	10/10 PASS
jrc_shelf_life_arrhenius	10	10/10 PASS
jrc_shelf_life_linear	16	16/16 PASS
jrc_shelf_life_poolability	10	10/10 PASS
jrc_shelf_life_extrapolate	11	11/11 PASS
TOTAL	57	57/57 PASS

5.1 jrc_shelf_life_q10 — 10/10 PASS

TC ID	Description	Expected	Result
TC-SHELF-Q10-001	Valid inputs	exit 0; key sections in output	**PASS**
TC-SHELF-Q10-002	Known AF = 8.0 (±0.001), real_time = 208.0 (±0.01)	Independent Python arithmetic reference	**PASS**
TC-SHELF-Q10-003	Sensitivity table rows Q10 −0.5, Q10, Q10+0.5	exit 0; '1.5', 'stated value', '2.5' in output	**PASS**
TC-SHELF-Q10-004	Q10=1.2 out-of-range warning	exit 0; range warning in output	**PASS**
TC-SHELF-Q10-005	accel_temp ≤ real_temp rejected	exit non-zero	**PASS**
TC-SHELF-Q10-006	No arguments	exit non-zero or usage	**PASS**
TC-SHELF-Q10-007	Non-numeric q10	exit non-zero	**PASS**
TC-SHELF-Q10-008	Negative accel_time	exit non-zero	**PASS**
TC-SHELF-Q10-009	--help flag	exit 0; 'Usage' in output	**PASS**
TC-SHELF-Q10-010	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	**PASS**

5.2 jrc_shelf_life_arrhenius — 10/10 PASS

TC ID	Description	Expected	Result

TC-SHELF-ARR-001	Valid inputs	exit 0; key sections in output	**PASS**
TC-SHELF-ARR-002	Known AF = 13.7826 (±0.001), real_time = 358.35 (±0.05)	Independent math.exp() reference	**PASS**
TC-SHELF-ARR-003	Kelvin inputs give same AF as Celsius	AF matches ±0.01	**PASS**
TC-SHELF-ARR-004	Sensitivity table Ea−2, Ea, Ea+2	exit 0; all three values in output	**PASS**
TC-SHELF-ARR-005	Ea=30 out-of-range warning	exit 0; range warning in output	**PASS**
TC-SHELF-ARR-006	accel_temp ≤ real_temp rejected	exit non-zero	**PASS**
TC-SHELF-ARR-007	No arguments	exit non-zero or usage	**PASS**
TC-SHELF-ARR-008	Invalid --unit F	exit non-zero	**PASS**
TC-SHELF-ARR-009	--help flag	exit 0; 'Usage' in output	**PASS**
TC-SHELF-ARR-010	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	**PASS**

5.3 jrc_shelf_life_linear — 16/16 PASS

TC ID	Description	Expected	Result
TC-SHELF-LIN-001	Valid 15-row dataset	exit 0; key sections in output	**PASS**
TC-SHELF-LIN-002	Known shelf life = 24.723 ±0.05	Independent Python OLS + bisection reference	**PASS**
TC-SHELF-LIN-003	Homogeneous variance — BF pass	exit 0; 'homogeneous' in output	**PASS**
TC-SHELF-LIN-004	Heterogeneous variance — warning, exit 0	warning in output	**PASS**
TC-SHELF-LIN-005	PNG saved to ~/Downloads/	*_jrc_shelf_life_linear.png found	**PASS**
TC-SHELF-LIN-006	Model CSV saved to ~/Downloads/	*_jrc_shelf_life_linear_model.csv found	**PASS**
TC-SHELF-LIN-007	--direction high	exit 0; 'upper' or 'high' in output	**PASS**
TC-SHELF-LIN-008	Values below spec at t=0 (direction=low)	exit non-zero	**PASS**
TC-SHELF-LIN-009	No arguments	exit non-zero or usage	**PASS**
TC-SHELF-LIN-010	Missing 'value' column	exit non-zero; 'value' in error	**PASS**
TC-SHELF-LIN-011	< 3 distinct time points	exit non-zero	**PASS**
TC-SHELF-LIN-012	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	**PASS**
TC-SHELF-LIN-013	--transform log — happy path	exit 0; 'log' in output	**PASS**
TC-SHELF-LIN-014	--transform log — known shelf life (±0.05)	Independent Python log-OLS + bisection reference	**PASS**
TC-SHELF-LIN-015	--transform log with value ≤ 0	exit non-zero; positive/<=0 in error	**PASS**
TC-SHELF-LIN-016	--transform log — model CSV transform field	model CSV contains transform = log	**PASS**

5.4 jrc_shelf_life_poolability — 10/10 PASS

TC ID	Description	Expected	Result
TC-SHELF-POOL-001	Valid 3-batch dataset	exit 0; 'Decision', 'POOL', Step 1/2 in output	**PASS**
TC-SHELF-POOL-002	FULL POOL decision; both ANCOVA steps p > 0.25	exit 0; 'FULL POOL'; p1 > 0.25; p2 > 0.25	**PASS**
TC-SHELF-POOL-003	DO NOT POOL; Step 1 p < 0.05	exit 0; 'DO NOT POOL'; Step 1 p < 0.05	**PASS**
TC-SHELF-POOL-004	PARTIAL POOL; Step 1 p > 0.25, Step 2 p < 0.01	exit 0; 'PARTIAL POOL'; p1 > 0.25; p2 < 0.01	**PASS**
TC-SHELF-POOL-005	PNG saved to ~/Downloads/	*_jrc_shelf_life_poolability.png found	**PASS**
TC-SHELF-POOL-	No arguments	exit non-zero or usage	**PASS**

006			
TC-SHELF-POOL-007	Missing 'batch' column	exit non-zero; 'batch' in error	**PASS**
TC-SHELF-POOL-008	Only 1 batch	exit non-zero	**PASS**
TC-SHELF-POOL-009	< 3 time points per batch	exit non-zero	**PASS**
TC-SHELF-POOL-010	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	**PASS**

5.5 jrc_shelf_life_extrapolate — 11/11 PASS

TC ID	Description	Expected	Result
TC-SHELF-EXT-001	Valid model CSV, target within range	exit 0; 'Projection', 'Fitted value'	**PASS**
TC-SHELF-EXT-002	Known fit = 84.502 (±0.01), CI lower = 83.976 (±0.05)	Closed-form from known coefficients + NIST t_crit	**PASS**
TC-SHELF-EXT-003	Target within range, above spec — PASS	exit 0; 'supported' in output	**PASS**
TC-SHELF-EXT-004	Target beyond shelf life — FAIL message	exit 0; 'NOT supported' or 'crossed'	**PASS**
TC-SHELF-EXT-005	>50% extrapolation warning	exit 0; warning in output	**PASS**
TC-SHELF-EXT-006	>100% extrapolation hard stop	exit non-zero; 'rejected' or '100' in output	**PASS**
TC-SHELF-EXT-007	No arguments	exit non-zero or usage	**PASS**
TC-SHELF-EXT-008	Missing model file	exit non-zero; 'not found' in output	**PASS**
TC-SHELF-EXT-009	Wrong-script model file	exit non-zero; 'script' in error	**PASS**
TC-SHELF-EXT-010	Negative target_time	exit non-zero	**PASS**
TC-SHELF-EXT-011	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	**PASS**

6. Deviations

No deviations were identified during OQ execution. All 57 test cases passed.

7. OQ Conclusion

The Operational Qualification of the JR Validated Environment Shelf Life & Degradation Analysis Module v1.0 is complete.

Result: PASS

All 57 automated test cases specified in JR-VP-SHELF-001 v1.0 were executed on 2026-04-19 and 57 passed. Results are derived from the OQ evidence file listed below.

- jrc_shelf_life_q10 — Accelerated Ageing via Q10 Method (ASTM F1980)
- jrc_shelf_life_arrhenius — Accelerated Ageing via Arrhenius Kinetics (ISO 11607 / ICH Q1E)
- jrc_shelf_life_linear — Linear Degradation Model with Shelf Life Estimation (ICH Q1E)
- jrc_shelf_life_poolability — ICH Q1E Batch Poolability Analysis (Two-Step ANCOVA)
- jrc_shelf_life_extrapolate — CI Projection to Target Time with Transform Support

Evidence of test execution is archived at: shelf_life_oq_execution_20260419T080311.txt

The Shelf Life & Degradation Analysis module is released as version 1.0 of repos/shelf_life/. Any subsequent change to the scripts, wrappers, or help files within repos/shelf_life/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.